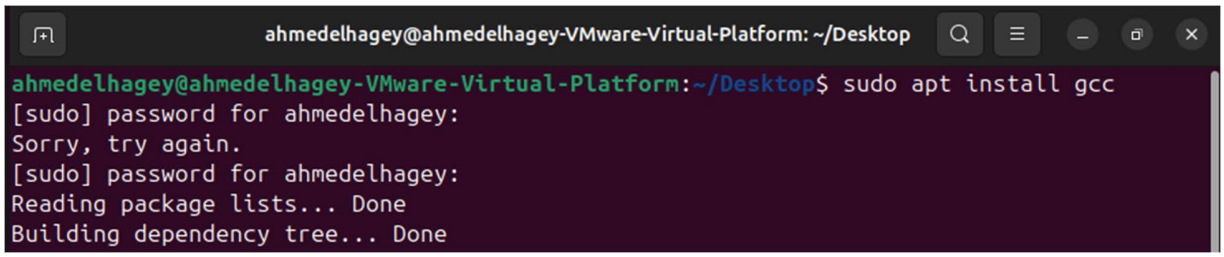
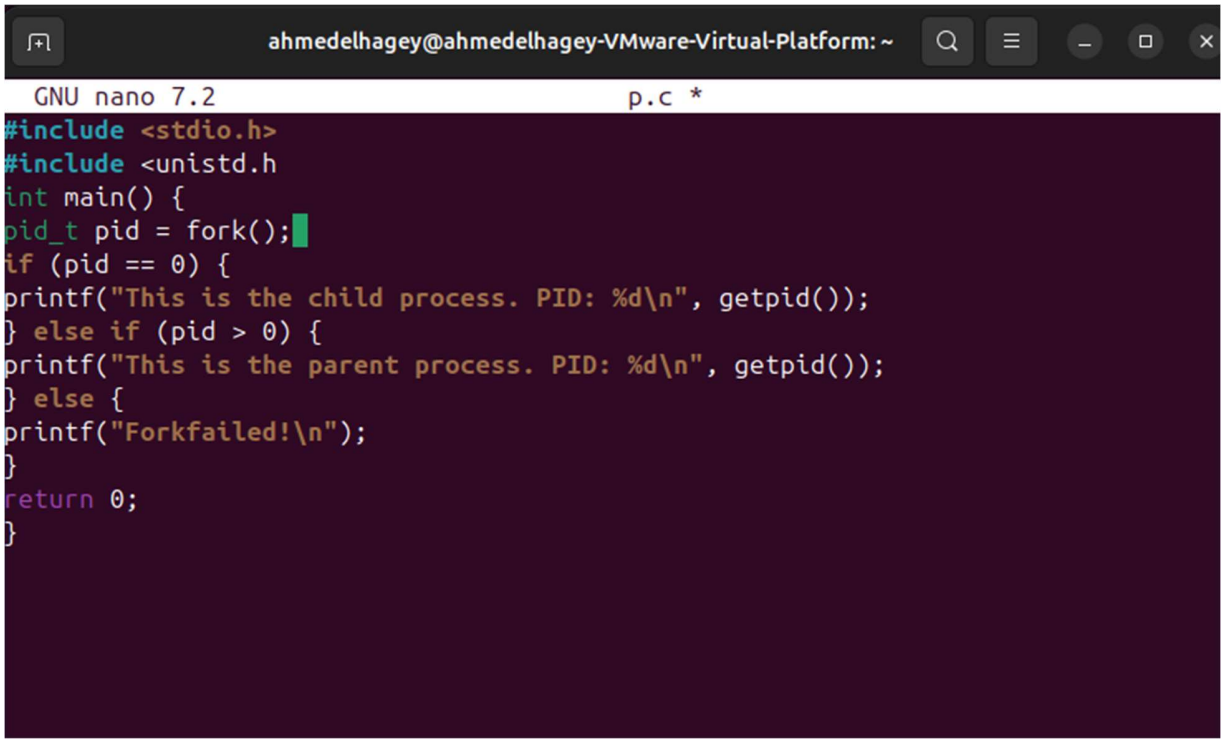
Assignment 2

Required Tools: GCC (GNU Compiler Collection): Required to compile C programs. Install with:
sudo apt install gcc



Task 1

Exercise 1: Using fork() in C Write a C program that demonstrates process creation using the fork() system call.



```
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ nano p.c
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ gcc p.c -o out
p.c:2:19: error: missing terminating > character
    2 | #include <unistd.h
      |                   ^
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ nano p.c
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ gcc p.c -o out
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ ./out
This is the parent process. PID: 6618
This is the child process. PID: 6619
```

Start and Stop a Process

Exercise 2: Starting Processes in the Background

Exercise 3: Stopping Processes

Exercise 4: Pausing and Resuming a Process

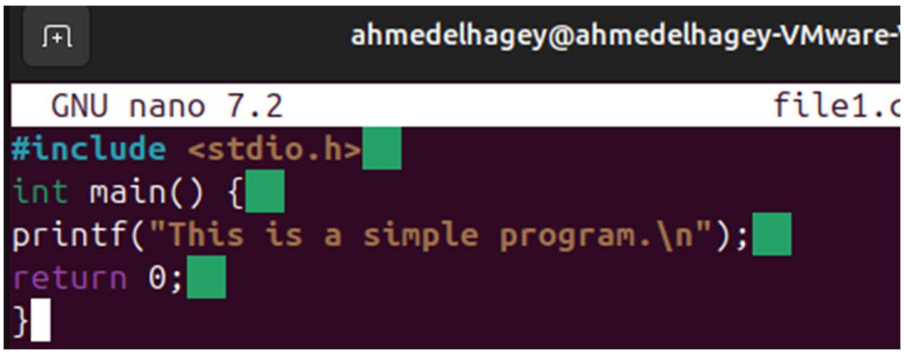
```
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ sleep 100 &
[1] 6625
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ sleep 300 &
[2] 6629
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ jobs
[1]-  Running                 sleep 100 &
[2]+  Running                 sleep 300 &
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ kill -STOP 6629
[1]   Done                    sleep 100

[2]+  Stopped                 sleep 300
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ jobs
                              sleep 300
Ubuntu 24.04.3 LTS amd64
                              hagey-VMware-Virtual-Platform:~$ kill -CONT 6629
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ jobs
[2]+  Running                 sleep 300 &
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ kill 6629
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ jobs
[2]+  Terminated              sleep 300
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$
```

```
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ gcc p.c -o out
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ ./out
This is the parent process. PID: 6755
This is the child process. PID: 6756
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ sleep 300 &
[1] 6757
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ jobs
[1]+  Running                 sleep 300 &
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ ps aux | grep sleep
ahmedel+    6757  0.0  0.0   8288  1940 pts/0    S    22:02   0:00 sleep 300
ahmedel+    6760  0.0  0.0   9144  2248 pts/0    S+   22:03   0:00 grep --color=
auto sleep
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ kill 6757
[1]+  Terminated              sleep 300
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$ ps aux | grep sleep
ahmedel+    6762  0.0  0.0   9144  2248 pts/0    S+   22:03   0:00 grep --color=
auto sleep
ahmedelhagey@ahmedelhagey-VMware-Virtual-Platform:~$
```

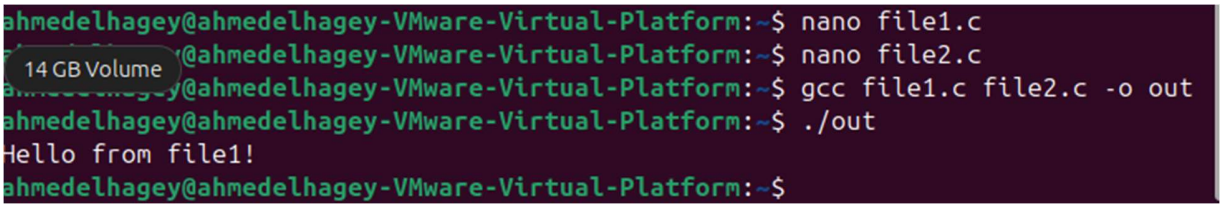
Task 2

Exercise 5: Role of the Linker

File 1.c

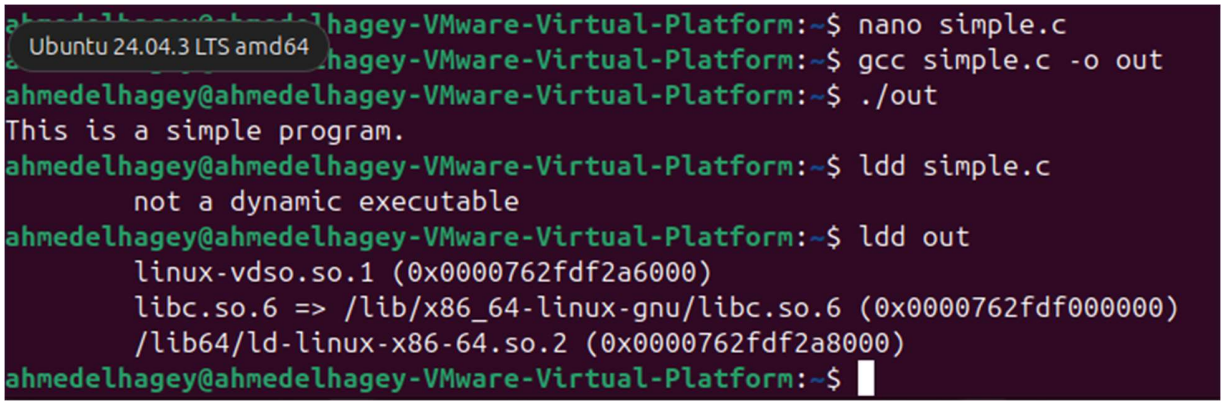


File2.c



Task 3

Exercise 6: Role of the Loader Write a simple program and inspect the libraries it uses with ldd Simple Program:



Make file

```
CC = gcc
CFLAGS = -Wall -Wextra -g

all: task1 task2 task3

task1: p.c
	$(CC) $(CFLAGS)    p.c  -o task1


Task2: file 1.c file2.c
	$(CC) $(CFLAGS)    file 1.c file2.c -o task2
Task3: simple.c
	$(CC) $(CFLAGS)    simple.c -o task3


clean:
	rm -rf  task1 task2 task3
```